Semantic Theory 2026: Practice Exam

Due* by Monday, June 29 11:59 am

You have 120 minutes to do this exam. The exam consists of 9 questions, of which the sub-questions (labeled: a, b, ...) are worth 10 points each (140 points total). In order to pass, you must get at least 70 points.

Please number every sheet of paper that you submit, and note the total number of sheets on the first page. You may not use any additional materials beyond those distributed together with this exam. Please do not use pencils.

Question 1: Predicate Logic (10)

a. Translate the following sentences into first-order predicate logic:

   i. *not all students <u>turn in</u> every assignment*

   ii. *Mary doesn't see everything that she hears*

Question 2: Type Theory (20)

Provide the derivations (type inferencing) of each of the following sentences. Brackets indicate the combinatorics, and subscripts indicate the types of (some of) the expressions—the rest must be deduced. You can treat *"the ship"* as the single term s' in (b).

a. [*some dog*] [$\mathit{ate}_{\langle e,\langle e,t\rangle\rangle}$[*a biscuit*]]

b. [*every gardener*] [*likes* $\underline{\mathit{potatoes}}_e$]

Question 3: λ-Calculus (10)

a. Given the types that you determined (or were given) for the terms in Question 2, derive the corresponding λ-expressions for the following terms:

   i. *likes*

   ii. *some*

   iii. *a*

Question 4: Generalized Quantifiers (20)

Consider the following sentence: *All bankers except for Mark and Sally will retire early*

a. Give the generalized quantifier definition of the noun phrase *"only George"*.

b. What are the monotonicity properties (left and right) of *only*? Show how you derived these properties.

Question 5: Event Semantics (20)

a. Translate the following sentences into Davidsonian (event semantics) representations, **including** temporal information.

   i. *a woman witnessed a robbery on Saturday*

   ii. $\underline{she}_e$ *she saw a burglar taking a watch*

b. For one of the sentences in (a) (you pick), draw a Davidsonian model structure in which the sentence holds. You may ignore the temporal aspects of the sentences here.

Question 6: Lexical Semantics (20)

Consider the following sentence: $S =$ *"two students ate a pizza"*

a. How many readings does the sentence S have? List all possible readings in natural language.

b. Translate each reading of S to the extended first-order logic for plural terms, where variables $\mathrm{X}, \mathrm{Y}, \mathrm{Z}, \ldots$ range over proper sums, $\mathrm{X} \oplus \mathrm{Y}$ denotes the group consisting of X and Y, $\lhd$ denotes the part-of-relation, and $N(\mathrm{X}) = |\{y \mid At(y) \wedge y \lhd \mathrm{X}\}|$ takes a proper sum X and returns the number of atoms in X.

Question 7: Dynamic Semantics (10)

Translate the following natural language utterances into Dynamic Predicate Logic..

i. *If* $\underline{Mary}_i$ *eats a sandwich,* $\underline{she}_i$ *is happy.* $\underline{She}_i$ *ate* $\underline{\text{a sandwich}}_k$. $\underline{It}_k$ *was delicious.*

ii. *Everyone who eats a sandwich drinks a beer.*

Question 8: DRT and Presuppositions (30)

a. Give proto-DRSs for the following sentences. Treat *"goes to"* in (i) as the two-place predicate *go-to*(x, y):

   i. *Every student finishes their homework or goes to the party*

   ii. *If Mary misses the bus, she is sad*

b. Resolve the proto-DRSs in (a). Explicitly describe the resolution constraints you apply.

c. Give the truth-conditions for one of the **resolved** DRSs in (b) (you pick). Use verifying embeddings to arrive at the model-theoretic interpretation.